

To show: $p \vee \neg p$	; 6 claim "excluded-middle"
proof	
Assume that $\neg p \vee \neg \neg p$	; 8 open block with assumption
Assume that p	; 9 open block with assumption
$\Rightarrow p \vee \neg p$	; 10 by 9, "or-intro"
$\Rightarrow \perp$	; 11 by (10, 8), "bottom-intro"
$\Rightarrow p \Rightarrow \perp$	; 12 by "impl-intro"
$\Rightarrow \neg p$	; 12 by "not-intro"
$\Rightarrow p \vee \neg p$	; 13 by 12, "or-intro"
$\Rightarrow \perp$	; 14 by (13, 8), "bottom-intro"
$\Rightarrow \neg p \vee \neg \neg p \Rightarrow \perp$	; 15 by "impl-intro"
$\Rightarrow \neg \neg p \vee \neg p$	; 15 by "not-intro"
$\Rightarrow p \vee \neg p$	; 15 by 15, "not-not"
qed	
$\Rightarrow var q$	; added variable
To show: $q \vee \neg q$	; 20 claim "excluded-middle"
proof	
Assume that $\neg q \vee \neg \neg q$	; 22 open block with assumption
Assume that q	; 23 open block with assumption
$\Rightarrow q \vee \neg q$	; 24 by 23
$\Rightarrow \perp$	; 25 by 23
$\Rightarrow q \Rightarrow \perp$	; 26 by "impl-intro"
$\Rightarrow \neg q$	; 26 by "not-intro"
$\Rightarrow q \vee \neg q$	; 27 by 26, "not-not"
$\Rightarrow \perp$	; 28 by 27, 26
$\Rightarrow \neg q \vee \neg \neg q \Rightarrow \perp$	; 29 by "impl-intro"
$\Rightarrow \neg \neg q \vee \neg q$	; 29 by "not-intro"
$\Rightarrow q \vee \neg q$	; 30 by 29, "not-not"
qed	
To show: $\%p \vee \neg \%p$	; 34 claim "excluded-middle"
proof	
Assume that $\neg \%p \vee \neg \neg \%p$	; 36 open block with assumption
Assume that $\%p$	; 37 open block with assumption
$\Rightarrow \%p \vee \neg \%p$	; 38 by 37
$\Rightarrow \perp$	; 39 by 37
$\Rightarrow \%p \Rightarrow \perp$	; 40 by "impl-intro"
$\Rightarrow \neg \%p$	; 40 by "not-intro"
$\Rightarrow \%p \vee \neg \%p$	; 41 by 20
$\Rightarrow \perp$	; 42 by 41, 40
$\Rightarrow \neg \%p \vee \neg \neg \%p \Rightarrow \perp$	; 43 by "impl-intro"
$\Rightarrow \neg \neg \%p \vee \neg \%p$	; 43 by "not-intro"
$\Rightarrow \%p \vee \neg \%p$	; 44 by 20
qed	